

baUSD Technical Architecture

A Soroban-native vault for real-world reinsurance yield

This document specifies the complete technical architecture of baUSD. The architecture is finalized at time of application; Tranche 1 is development, not design. The protocol is built end-to-end on the Stellar stack and composes with existing ecosystem building blocks for distribution, wallet access, liquidity, and cross-chain inflow.

1 Overview

baUSD is a yield-bearing token issued natively on Soroban. An allowlisted institutional LP deposits USDC on Stellar into a vault contract and mints baUSD, an appreciating share token whose value rises as earned premium from regulated US casualty specialty reinsurance treaties accrues. Yield is uncorrelated to crypto, equities, and interest rates because it is sourced from insurance underwriting profit, not token emissions, lending spread, or duration.

The defining design constraint is that **deposited capital does not remain in the contract**. The majority flows off-chain into reinsurance treaties through a Bermuda Class 3A Segregated Accounts Company (SAC) and is collateralized in Reg 114 Trusts. Consequently the vault holds only a liquidity sleeve for near-term redemptions, the on-chain `total_assets` is an attested NAV rather than a balance of held USDC, and redemptions are gated by an allowlist and a notice period because casualty tail development means capital cannot be returned on demand at full value. This is what separates baUSD from a synthetic or wrapped representation: the share token tracks attested treaty performance, including downside.

2 Stellar Relevance — Core, Not Superficial

Stellar is not used for storage or as a thin wrapper. The core financial logic, share accounting, mint/burn, NAV accrual, and redemption, executes inside Soroban smart contracts. baUSD exists only as a Stellar asset, its yield-bearing behavior is enforced on-chain, and its distribution runs through Stellar-native yield infrastructure. Remove Stellar and there is no product. The ecosystem building blocks in section 6 are leverage on top of this protocol, not the protocol itself.

3 System Architecture

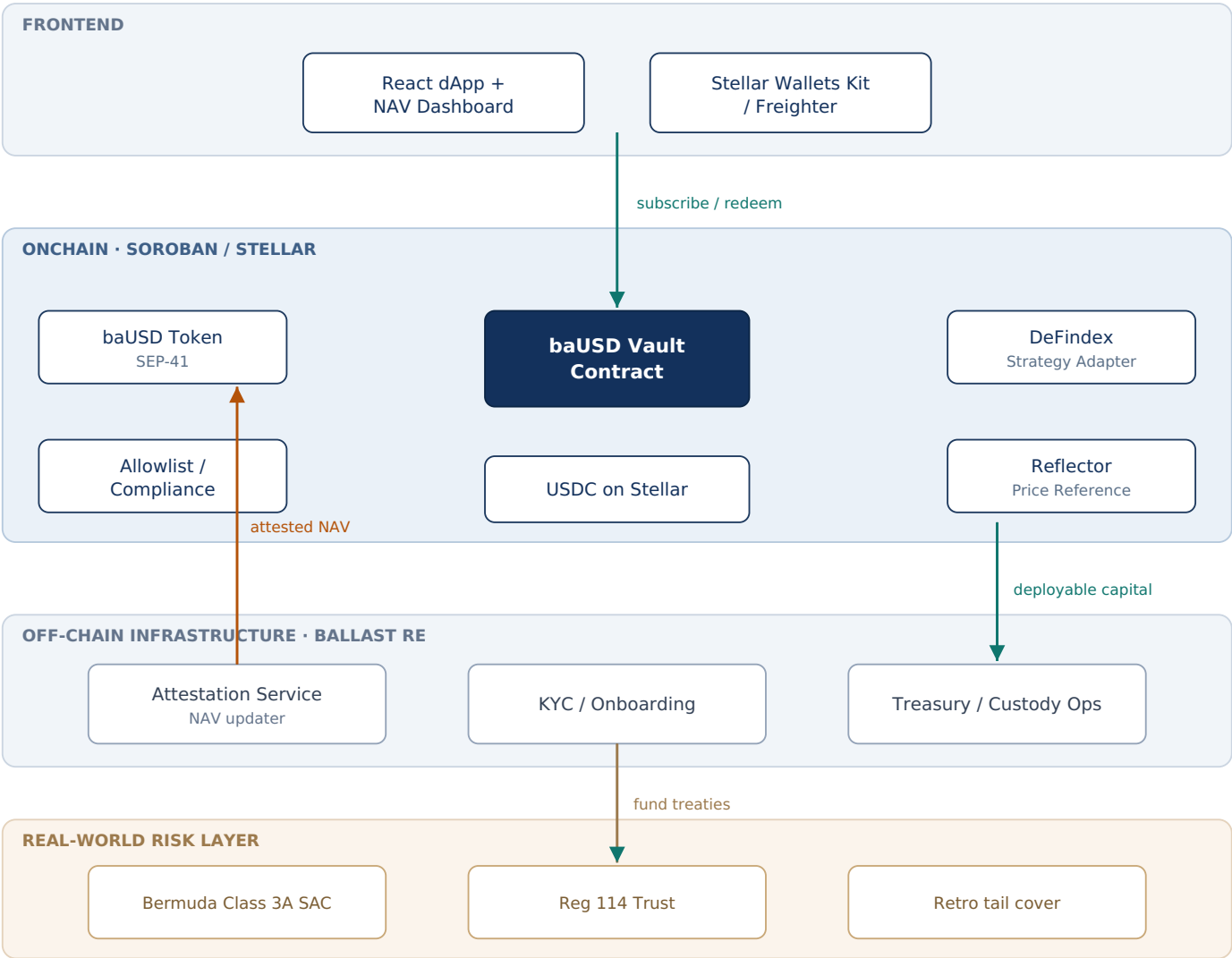


Figure 1. System architecture. Capital flows down to the real-world risk layer; attested NAV flows back up to the vault.

Layer	Components	Responsibility
Frontend	React dApp + dashboard, Stellar Wallets Kit / Freighter	Subscribe, redeem, view NAV/position
Onchain	Vault contract, baUSD token (SEP-41), allowlist module, Reflector reference, DeFindex adapter	Share accounting, mint/burn, NAV state, compliance gating, composability
Off-chain infra	Attestation service, KYC service, treasury ops	NAV sourcing/attestation, identity verification, capital movement
Real-world risk	Bermuda SAC, Reg 114 Trust, retro cover	Underwriting, loss reserving, tail protection

3.1 baUSD token contract (SEP-41)

baUSD is implemented as a SEP-41-compatible token contract. Mint and burn authority is held exclusively by the vault contract (set as token admin); there is no discretionary issuance path, so supply is bound 1:1 to vault subscriptions and redemptions. `transfer` is gated by the allowlist check (see 3.3); `balance` and `allowance` are standard.

3.2 Vault contract (appreciating-share / ERC-4626-equivalent)

The vault is the core logic contract. baUSD shares represent a pro-rata claim on the attested NAV; $\text{share value} = \text{total assets} / \text{total shares}$. There is no rebasing, balances are constant and value-per-share moves.

```
shares_minted    = deposit_usdc * total_shares / total_assets    // at subscription
assets_returned  = shares_burned * total_assets / total_shares  // at redemption
share_price      = total_assets / total_shares                  // NAV per baUSD
```

On first deposit, `share_price` initializes to 1.0 (1 baUSD = 1 USDC).

State variable	Meaning
total_shares	total baUSD outstanding
total_assets	attested NAV in USDC terms (off-chain deployed + onchain sleeve)
liquidity_sleeve	USDC actually held onchain for redemptions
nav_last_updated	ledger timestamp of last attestation
attestation_authority	multisig permitted to update NAV
compliance_authority	key/multisig permitted to manage allowlist
admin	governance multisig
paused	guardian-controlled circuit breaker
redemption_notice	configurable notice/lockup period

Core entrypoints

- `subscribe(amount)` — allowlist-checked; pulls USDC, mints baUSD at current share price
- `request_redemption(shares)` — allowlist-checked; registers a redemption request and starts the notice period
- `claim_redemption(request_id)` — after notice elapses, burns baUSD and pays USDC from the sleeve at NAV
- `update_nav(new_total_assets, proof_ref)` — attestation-authority only; constrained (see 7.2)
- `fund_sleeve(amount)` / `deploy_capital(amount)` — treasury ops; moves USDC between onchain sleeve and off-chain deployment, logged as events
- `set_allowlist_authority`, `set_attestation_authority`, `pause`, `unpause`, `upgrade` — governance/guardian

3.3 Allowlist / compliance module

A module maintains the set of KYC-verified institutional addresses. `subscribe` and `request_redemption` revert for non-allowlisted callers, and baUSD `transfer` checks both sender and recipient, preventing secondary distribution to unverified parties. The compliance authority adds/removes addresses based on off-chain KYC completion.

3.4 NAV / yield-accrual module

An authorized attestation service raises (or lowers) on-chain NAV from off-chain attested earned-premium and reserve data, driving baUSD appreciation. The update path is constrained on-chain (section 7.2) so the largest trust assumption in the system has a bounded blast radius.

4 Capital & Value Flow



baUSD is not pegged; adverse loss development can reduce NAV and the share price.

Figure 2. Capital lifecycle from onboarding through redemption.

- **Deposit:** an allowlisted LP sends USDC to the vault; the vault mints baUSD at the current share price.
- **Deployment:** USDC beyond the sleeve is applied off-chain to bound casualty treaties via the Bermuda SAC reinsurer, recorded against on-chain NAV.
- **Accrual:** as treaties earn premium, attested amounts update on-chain NAV and baUSD share value rises.
- **Redemption:** the holder requests redemption, the notice period elapses, then claims; the vault burns baUSD and releases USDC at NAV from the sleeve. Requests beyond available sleeve liquidity queue until it is refunded from off-chain, reflecting reinsurance tail timelines.
- **Downside:** adverse loss development reduces attested `total_assets`, so `update_nav` can lower the share price. baUSD is not pegged and does not guarantee principal; it tracks treaty performance behind the collateral waterfall.

5 Yield Mechanism

baUSD uses an appreciating-share model, not rebasing: total vault assets grow as premium is earned, so each baUSD redeems for more USDC over time. The 6%+ APY target is net earned reinsurance premium, not token emissions, lending spread, or T-bill duration, which is the source of its non-correlation to crypto and rates.

6 Ecosystem Composability & Distribution

baUSD composes with existing Stellar building blocks rather than rebuilding them. Funded items sit inside the \$85K award scope (Tranche 2, testnet/feature-complete); roadmap items are post-mainnet composability expansions, scoped with their published integration times and deliberately kept out of award scope so the budget stays matched to team and timeline.

Building block	Category	How baUSD composes	Est. time	Status
DeFindex (Palta Labs)	DeFi / Yield	baUSD published as a DeFindex Strategy conforming to the strategy interface, so any DeFindex-integrated wallet or app can allocate into baUSD and surface reinsurance yield.	1-2 weeks	Funded · Tranche 2
Stellar Wallets Kit	Wallet	Multi-wallet connect layer in the dApp (Freighter, xBull, Albedo) for a single auth flow.	Under 1 day	Funded · Tranche 2
Freighter Connect (SDF)	Wallet	Primary Soroban browser wallet for direct vault interaction.	1-2 weeks	Funded · Tranche 2
Blend v2	DeFi / Lending	baUSD onboarded as a collateral asset so holders can borrow against reinsurance yield without exiting the position.	1+ month	Roadmap · post-mainnet
Soroswap (Palta Labs)	DeFi / Liquidity	baUSD/USDC pair plus routing aggregating Soroban and classic liquidity.	1-2 weeks	Roadmap
Aquarius	DeFi / Liquidity	Liquidity aggregation and market data for the baUSD pair.	Under 1 day	Roadmap
CCTP (Circle)	Cross-chain	Native 1:1 USDC inflow to Stellar from other chains, letting LPs holding USDC elsewhere deposit without third-party bridges.	1-5 days	Roadmap

Primary distribution — DeFindex (funded)

DeFindex is Stellar's yield infrastructure for wallets and DeFi apps. baUSD is implemented as a DeFindex Strategy conforming to the strategy interface, validated against the DeFindex Quickstart and API Reference, with developer support via the Palta Labs Discord. This turns baUSD from a standalone vault into wallet-distributable yield: any DeFindex-integrated wallet can offer reinsurance yield without bespoke work. The adapter holds no custody logic of its own; it calls the vault's standard entrypoints.

Wallet access (funded)

The dApp uses Stellar Wallets Kit for a unified multi-wallet connect flow and supports Freighter Connect as the primary Soroban browser wallet for direct vault interaction. Both are quick integrations included in the Tranche 2 dApp deliverable.

Lending, liquidity, cross-chain (roadmap)

Post-mainnet, baUSD is onboarded as Blend v2 collateral (borrow against the position without exiting), paired on Soroswap with Aquarius market data for secondary entry/exit before the development period completes, and made fundable cross-chain via CCTP, widening the addressable capital base while keeping settlement on Stellar.

Oracle / price reference

baUSD's NAV is set on-chain by the constrained attestation authority. For external, verifiable price discovery used by downstream integrators, baUSD's share price is published via Reflector, the de-facto Stellar oracle. Oracles are not a category on the SCF Integration List; Reflector is noted for completeness, not claimed as a listed building block.

7 Security, Risk & Trust Model

7.1 Roles and separation of duties

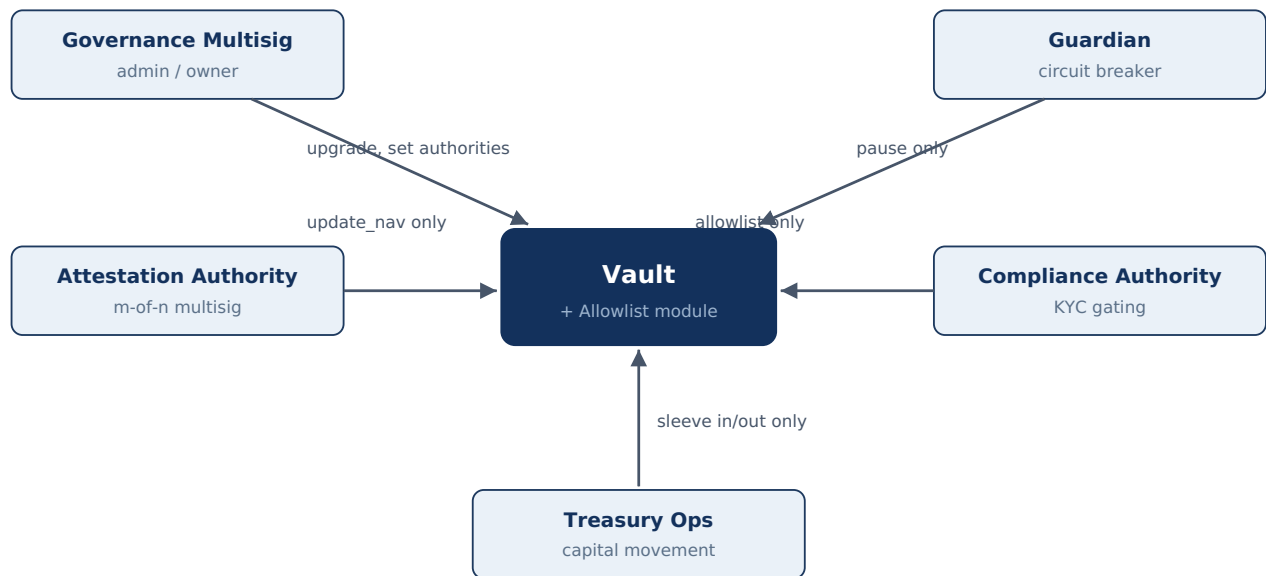


Figure 3. Each authority is scoped to the minimum surface it needs. Attestation can move NAV but not funds; treasury can move the sleeve but not NAV; compliance gates addresses but touches neither.

The attestation authority can move NAV but cannot move funds; treasury ops can move the sleeve but cannot set NAV; compliance can gate addresses but touches neither funds nor NAV.

7.2 NAV attestation constraints

The attested NAV is the single largest trust assumption, constrained on-chain to limit blast radius:

- **Bounded delta:** `update_nav` reverts if the change exceeds a configurable per-update cap, forcing large moves to be staged and visible.
- **Multisig:** attestation authority is an m-of-n multisig, not a single key.
- **Freshness:** updates carry a timestamp; consumers and Reflector can treat NAV beyond a max age as stale/invalid.
- **Proof reference:** each update logs a `proof_ref` (hash/URI of the supporting earned-premium and reserve attestation) for auditability.
- **Monotonicity not assumed:** NAV can decrease. Downward updates are explicitly permitted so the token reflects real losses rather than masking them.

7.3 Other controls

- **Smart-contract security:** independent audit via the SCF Audit Bank (not in budget), with remediation before mainnet.
- **Access control:** vault-bound mint/burn authority; allowlist-gated subscription and redemption; Soroban `require_auth` on all privileged calls.
- **Reentrancy:** checks-effects-interactions ordering; state updated before external token transfers.
- **Pausability:** guardian can pause subscribe/redeem in an incident without touching balances.
- **Upgradeability:** Soroban contract upgrade gated behind the governance multisig and a configurable timelock, giving holders advance notice of code changes.
- **Liquidity risk:** the redemption notice period and sleeve sizing are the buffer against a redemption rush exceeding onchain liquidity; excess requests queue until the sleeve is refunded.

7.4 Off-chain trust assumptions (explicit)

Disclosed rather than engineered away: accuracy of earned-premium and reserve data feeding attestation (mitigated by independent actuarial review); custody and movement of off-chain capital by treasury ops; solvency and claims-paying behavior of the SAC, backstopped by the collateral waterfall (earned premium reserves, Reg 114 Trust loss reserves, segregated-account structure, retrocessional tail cover).

8 Compliance & Proof of Reserves

Institutional LPs are KYC/AML-verified off-chain; verified addresses are allowlisted on-chain for subscription and redemption. The Bermuda SAC / BVI structure provides the regulated wrapper for the underlying reinsurance activity. A Stellar Info File (SEP-1 stellar.toml) will be published for the baUSD asset linking to third-party proof of backing, the independent actuarial review plus Reg 114 Trust reserve confirmation, prior to mainnet.

9 Deployment & Open Source

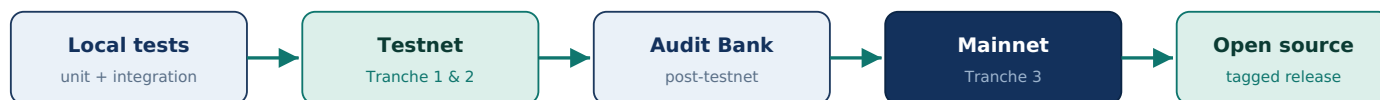


Figure 4. Deployment pipeline. Apache-2.0 contracts are public from testnet; tagged with deployment addresses at mainnet.

- **MVP (Tranche 1):** vault + token + tests on testnet; deposit, mint, redeem demo.
- **Testnet / feature complete (Tranche 2):** NAV attestation, allowlist gating, Reflector reference, DeFindex strategy, Stellar Wallets Kit + Freightier dApp.
- **Mainnet (Tranche 3):** verified deployment, public NAV API + dashboard + monitoring/alerting, developer docs, first real USDC deposit.

Open source: all Soroban contracts released under Apache-2.0, public from testnet (Tranche 2), fully tagged and documented with deployment addresses at mainnet (Tranche 3). Off-chain underwriting models, attestation data pipelines, and proprietary risk logic remain proprietary and outside the published scope.

10 Why Stellar

- Low, predictable fees suited to frequent NAV updates and institutional settlement.
- Leading tokenized-RWA ecosystem (Franklin Templeton BENJI, Ondo USDY, WisdomTree); reinsurance premium is the logical next RWA yield vertical, yielding more than tokenized treasuries.
- DeFindex distribution turns baUSD into productive, wallet-accessible yield rather than idle TVL.
- A deep set of composable building blocks (Blend, Soroswap, Aquarius, CCTP) lets baUSD become collateral, liquidity, and cross-chain-fundable over time.

A Appendix — Configurable Parameters to Finalize

Set to real values before submission/build:

1. Redemption notice / lockup period (calibrated to casualty tail development).
2. Onchain liquidity sleeve size as a percentage of NAV.
3. NAV per-update delta cap and update cadence.
4. Attestation authority multisig threshold (m-of-n).
5. Upgrade timelock duration.
6. Whether baUSD transfers are allowlist-gated on both sides or subscription-only.